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- **Overview**

| <b>Option / data</b> | <b>Original data</b> | <b>Undersampling</b> | <b>rosrus</b> |
|----------------------|----------------------|----------------------|---------------|
| <b>Count Vec</b>     | Increase             | Decrease             | Decrease      |
| <b>TF-IDF Vec</b>    | No effect            | Increase             | Decrease      |
| <b>Stem</b>          | No effect            | Increase             | Increase      |
| <b>Lemma</b>         | No effect            | Decrease             | Decrease      |
| <b>SW Kept</b>       | Decrease             | Increase             | Increase      |
| <b>SW removed</b>    | Increase             | Decrease             | Decrease      |

**Original data : all info, high bias**

**undersampling : lose info, lower bias**

**rosrus : worst between them**

# Original data

- The experiment :
  - BoW (CountVectorizer), TF-IDF
  - Stemming , Lemmatization
  - With/without Stopwords
- models:
  - SVM
  - Multinomial Naive Bayes
  - MLP
  - HMM
- MLP/HMM is tested with dimension reduced data (TruncatedSVD)

| SVC (with classwight) | Accuracy | Recall | Precision | F1-score |
|-----------------------|----------|--------|-----------|----------|
| BOW                   | 0.661    | 0.70   | 0.62      | 0.65     |
| POS                   | 0.17     | 0.24   | 0.19      | 0.16     |
| NER                   | 0.18     | 0.19   | 0.18      | 0.12     |
| POS+NER               | 0.20     | 0.27   | 0.21      | 0.19     |

| SVC<br>noClassWiegth<br>Stem<br>BoW | Accuracy | Recall | Precision | F1-score |
|-------------------------------------|----------|--------|-----------|----------|
| BOW                                 | 0.70     | 0.53   | 0.77      | 0.60     |

| Multinom<br>ial Naive<br>Bayes | Accuracy | Recall | Precision | F1-score |
|--------------------------------|----------|--------|-----------|----------|
| BOW                            | 0.67     | 0.50   | 0.73      | 0.56     |
| POS                            | 0.41     | 0.14   | 0.16      | 0.14     |
| NER                            | 0.43     | 0.12   | 0.17      | 0.11     |
| POS+NER                        | 0.35     | 0.18   | 0.18      | 0.17     |

| MLP         | Accuracy | Recall | Precision | F1-score |
|-------------|----------|--------|-----------|----------|
| BOW         | 0.59     | 0.33   | 0.64      | 0.38     |
| POS         | 0.46     | 0.12   | 0.27      | 0.10     |
| NER         | 0.46     | 0.11   | 0.13      | 0.08     |
| POS+N<br>ER | 0.47     | 0.14   | 0.39      | 0.14     |

| HMM StopWords<br>removed Stem<br>TF-IDF/BoW | Accuracy |
|---------------------------------------------|----------|
| BOW                                         | 0.085    |
| POS                                         | 0.081    |
| NER                                         | 0.055    |
| POS+NER                                     | 0.169    |

# Under sampling the high classes

- applying undersampling on large classes (10,000) to be close to small classes on range(9,000–5,000)
- The experiment:
  - BoW (CountVectorizer) , TF-IDF
  - Stemming , Lemmatization
  - With/without Stopwords
- models:
  - SVM
  - Multinomial Naive Bayes
  - MLP
  - HMM
- MLP/HMM is tested with dimension reduced data (TruncatedSVD)

| SVC (with classwight) | Accuracy | Recall | Precision | F1-score |
|-----------------------|----------|--------|-----------|----------|
| BOW                   | 0.64     | 0.62   | 0.67      | 0.64     |
| POS                   | 0.26     | 0.23   | 0.24      | 0.20     |
| NER                   | 0.234    | 0.20   | 0.20      | 0.14     |
| POS+NER               | 0.29     | 0.25   | 0.29      | 0.23     |

| MLP     | Accuracy | Recall | Precision | F1-score |
|---------|----------|--------|-----------|----------|
| BOW     | 0.63     | –      | –         | –        |
| POS     | 0.25     | 0.22   | 0.23      | 0.19     |
| NER     | 0.233    | 0.20   | 0.20      | 0.14     |
| POS+NER | 0.28     | 0.25   | 0.26      | 0.23     |

| Multinomial Naive Bayes | Accuracy | Recall | Precision | F1-score |
|-------------------------|----------|--------|-----------|----------|
| BOW                     | 0.60     | 0.60   | 0.60      | 0.59     |
| POS                     | 0.20     | 0.18   | 0.17      | 0.16     |
| NER                     | 0.16     | 0.14   | 0.18      | 0.10     |
| POS+NER                 | 0.23     | 0.20   | 0.20      | 0.18     |

| HMM StopWords removed Stem TF-IDF/BoW | Accuracy |
|---------------------------------------|----------|
| BOW                                   | 0.1198   |
| POS                                   | 0.118    |
| NER                                   | 0.04     |
| POS+NER                               | 0.1195   |

# BOW – rosrus

- Data Balancing Strategy (ROS–RUS)  
To address the severe class imbalance in the original dataset, a hybrid resampling strategy was applied, combining Random OverSampling (ROS) and Random UnderSampling (RUS).
  - Avoids oversampling of minority classes (which may cause overfitting).
  - Avoids undersampling of majority classes (which may discard useful information).
- The experiment :
  - Bag of Words (BoW) using CountVectorizer was used only
  - Stemming , Lemmatization
  - With/without Stopwords
- models:
  - SVM
  - Multinomial Naïve Bayes
  - MLP
  - HMM
- MLP/HMM is tested with dimension reduced data (TruncatedSVD)

**Compared** to undersampling-only and original data, **ROS–RUS is the weakest balancing strategy** for this task

| SVC (with classweight) | Accuracy | Recall | Precision | F1-score |
|------------------------|----------|--------|-----------|----------|
| BOW                    | 0.47     | 0.70   | 0.51      | 0.52     |
| POS                    | 0.17     | 0.24   | 0.19      | 0.16     |
| NER                    | 0.18     | 0.20   | 0.17      | 0.12     |
| POS+NER                | 0.20     | 0.27   | 0.21      | 0.19     |

| MLP     | Accuracy | Recall | Precision | F1-score |
|---------|----------|--------|-----------|----------|
| BOW     | 0.368    | –      | –         | –        |
| POS     | 0.12     | 0.17   | 0.15      | 0.11     |
| NER     | 0.17     | 0.19   | 0.19      | 0.12     |
| POS+NER | 0.17     | 0.25   | 0.20      | 0.17     |

| Multinomial Naive Bayes | Accuracy | Recall | Precision | F1-score |
|-------------------------|----------|--------|-----------|----------|
| BOW                     | 0.56     | 0.72   | 0.50      | 0.56     |
| POS                     | 0.15     | 0.22   | 0.17      | 0.15     |
| NER                     | 0.13     | 0.13   | 0.19      | 0.09     |
| POS+NER                 | 0.15     | 0.20   | 0.17      | 0.14     |

| HMM StopWords removed Stem TF-IDF/BoW | Accuracy |
|---------------------------------------|----------|
| BOW                                   | 0.169    |
| POS                                   | 0.095    |
| NER                                   | 0.091    |
| POS+NER                               | 0.147    |